

Solution to Problem set 5: International Trade II

C.Cooperation

1. Explain what is the “prisoner’s dilemma” and give an example (unrelated to prisoners) where the incentive structure of two parties is similar to a prisoner’s dilemma.

A prisoner’s dilemma, based on the motivating story of two prisoners who must choose whether or not to confess, describes a strategic interaction where two players have a dominant strategy of choosing non-cooperative behavior. Consequently, the equilibrium is a non-cooperative one, which is however Pareto inefficient (as cooperation is assumed to be Pareto efficient). A common example is the nuclear arms race: each country thinks that if the other country does not invest in arms, it is better to invest in arms (and also if the other country does invest, it is better to invest to maintain a balance). Consequently, both waste money and potentially cause harm if the arms are deployed.

2. Explain how a bilateral international treaty that limits tariffs can be sustained, even if the choice of tariffs is similar to a prisoner’s dilemma.

As discussed in class, there are (at least) two theoretical concepts that support cooperation. First, “tacit collusion” – if the parties view the tariff decision as an infinitely repeated prisoners’ dilemma, they can choose a grim trigger strategy which leads to cooperation (as long as the payoff structure supports this, e.g. when the discount factors are not too high).

Second, according to cooperative game theory (the Coase Theorem), the parties will realize that there is a surplus to be made by collaboration and thus be able to reach an agreement as long as transaction costs are sufficiently low.

3. John is studying for his exam in the course on “Economics & Law”. Jill is his neighbor and listens to loud music that disturbs John’s concentration. You are the landlord and must choose between two options:
 - a. Option 1: John can force Jill to stop the noise.
 - b. Option 2: Jill can keep on making noise.

From an efficiency perspective, does it matter which option you choose? What does it depend on? (in your answer, please refer to the Coase Theorem).

The Coase Theorem states that it does not matter for efficiency who has the initial property right, as long as transaction costs are low. Here, if John values his right more than Jill, he will agree to pay her for stopping the noise and vice versa. This only works, however, if property rights are well-defined and transaction costs are low.

4. Alice declares herself the owner of the “deep water” in the ocean. Bob wants to spill chemicals in the water. He claims that the chemicals only stay in the “shallow water” and that Alice has no right to stop him. Alice claims that she not only has a right, but that she values her right by 1M\$. Bob claims that he values his right by 2M\$. Bob sues Alice. Alice says that the court should not intervene, because the Coase Theorem says that the parties should just negotiate and then they will reach an efficient outcome anyway. Is Alice’s correct?

Alice is incorrect. A precondition for the Coase Theorem is that rights are well-defined, and here “the deep water” is very vague. Hence, the parties cannot negotiate effectively. An alternative way to think about this is that determining what is “the deep water” is very costly, which means high transaction costs.

5. Country A and Country B sign a free trade agreement in which they each obligate not to impose any tariffs. The two countries have the following payoff structure:

Country A / Country B	Taarif	No Taarif
Taarif	2, 2	10, 0
No Taarif	0, 10	7, 7

- Do they countries have a dominant strategy? If so, what is it?
This is a prisoner’s dilemma: the dominant strategy is “taarif”.
- What is the (nash) equilibrium of the game if it is played one time?
(Taarif, Taarif).
- What is the (nash) equilibrium of the game if it is played 10 times?
The same. A finite number of repetitions do not change the result.
- Suppose that both countries have the same discount factor, of $= \frac{1}{2}$. What are then the (nash) equilibria of the game if it is played an *infinite amount of time* times?

See the answer below.

- What is the critical discount factor for which the countries might prefer to cooperate by playing a grim trigger strategy (where each countries avoids charging a Taarif as long as the other country does so as well, and switches to charging a taarif forever once the other country charges a taarif).

Hint: you can use the formula discussed in class:

$$\delta \geq \frac{\pi^D - \pi^C}{\pi^D - \pi^P}$$

Consider the formula $\delta \geq \frac{\pi^D - \pi^C}{\pi^D - \pi^P}$. The payoffs are:

$$\pi_D = 10, \pi_C = 7, \pi_P = 2$$

The critical discount factor is thus $= \frac{10-7}{10-2} = \frac{3}{8} < \frac{1}{2}$. Hence, with a discount factor of $\frac{1}{2}$ there can be cooperation and the critical discount factor is $\frac{3}{8}$.

D.Classifying goods

6. Which one of these statements is *incorrect*?
- a. A public good is non-rivalrous and non-excludible
 - b. A private good is rivalrous and excludible
 - c. A common good is non-rivalrous and excludible**
 - d. A club good is non-rivalrous and excludible

7. Explain why private goods should be privately owned whereas public goods should be publicly owned, from an efficiency perspective.

*The answer appears explicitly in the slides. *

8. Jane and John want to swim in the *Binnen Alster* (next to Jungfernstieg in Hamburg), but when they arrive there – they see that Max has sealed off the entire area. John files a petition to the court, saying Max should not be allowed to seal-off the area, because it is a public good. Is John correct?

There are several questions here (e.g. is Max the right owner) but from the perspective of classifying goods, we should ask whether the Binnen Alster is a public good. It seems mostly non-rivalrous (unless many many people jump in at the same time) but the fact that Max could seal it off suggests it is excludible. Hence, it is a “club good”.

9. Bill wants to buy a house that is owned by the government. The president objects to the sale, saying “This house is a public good, because it is owned by the government”. Is the president correct?

Bill is incorrect. For classification, it does not matter who the current owner is – that is just a factual question. What matters are the attributes of the good – is it rivalrous and is it excludible. Typically, a house is both – hence, a private good.

10. In the land of “Fantasialand”, the government owns the rights to the sky above the country. However, the neighboring countries disregard this right and fly airplanes above Fantasialand without approval. The president of Fantasialand is concerned that

there is an incentive problem causing the government to under-invest in keeping the sky free of foreign planes. Hence, he decides to give the rights to Michael, a private person. Are the skies now a private good? Is it a good idea to give rights to Michael?

The skies do not become a private good, just because Michael is technically allowed to exclude other. It is probably still too costly to exclude others, and hence not a private good. Defining whether the sky is a public good or common good is a bit tricky, because they have different features, e.g. from the perspective of “looking at the sky and enjoying the nice blue color”, it is non-rivalrous. But as routes for airplanes, they are rivalrous, and hence a common good.

11. The king decides to build a wall around a small lake, which was previously operated in an “open-access” way (i.e. anyone could get in). Does this change the classification of the lake?

As the king managed to exclude others, one might argue that this has been an excludable good all along. But of course, in some cases, it is possible to convert a public/common good by changing its features.